

HW2

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Section 1.3 #27

Find all 2×2 matrices of the form

$$A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$$

such that

$$A^2 = AA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

solving A^2 ,

$$A^2 = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} = \begin{pmatrix} a^2 & b(a+c) \\ 0 & c^2 \end{pmatrix}.$$

set this answer equal to I

$$a^2 = 1, \quad c^2 = 1, \quad b(a+c) = 0.$$

the first two equations give us $a = \pm 1$ and $c = \pm 1$. and for the third equation, if $a = c$ then $a + c \neq 0$, so $b = 0$; if $a \neq c$ then $a + c = 0$ automatically, so b has got to be any real number. this gives all solutions:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & b \\ 0 & -1 \end{pmatrix}, \quad b \in \mathbb{R}, \quad A = \begin{pmatrix} -1 & b \\ 0 & 1 \end{pmatrix}, \quad b \in \mathbb{R}.$$

Section 1.4 #5

find A^{-1} or show it does not exist, where

$$A = \begin{pmatrix} 0 & 1 & -1 \\ 3 & 1 & 1 \\ 1 & 2 & -1 \end{pmatrix}.$$

find A^{-1} .

$$\begin{aligned}
 & \left[\begin{array}{ccc|ccc} 0 & 1 & -1 & 1 & 0 & 0 \\ 3 & 1 & 1 & 0 & 1 & 0 \\ 1 & 2 & -1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 0 & 0 & 1 \\ 3 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 \end{array} \right] \\
 & \xrightarrow{R_2 - 3R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 0 & 0 & 1 \\ 0 & -5 & 4 & 0 & 1 & -3 \\ 0 & 1 & -1 & 1 & 0 & 0 \end{array} \right] \xrightarrow{R_1 - 2R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & -2 & 0 & 1 \\ 0 & -5 & 4 & 0 & 1 & -3 \\ 0 & 1 & -1 & 1 & 0 & 0 \end{array} \right] \\
 & \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & -2 & 0 & 1 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & -5 & 4 & 0 & 1 & -3 \end{array} \right] \xrightarrow{R_3 + 5R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & -2 & 0 & 1 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 5 & 1 & -3 \end{array} \right] \\
 & \xrightarrow{R_1 + R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 1 & -2 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 5 & 1 & -3 \end{array} \right] \xrightarrow{(-1)R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 1 & -2 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -5 & -1 & 3 \end{array} \right] \\
 & \xrightarrow{R_2 + R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 1 & -2 \\ 0 & 1 & 0 & -4 & -1 & 3 \\ 0 & 0 & 1 & -5 & -1 & 3 \end{array} \right]
 \end{aligned}$$

the left block is I , so the inverse exists and

$$A^{-1} = \begin{pmatrix} 3 & 1 & -2 \\ -4 & -1 & 3 \\ -5 & -1 & 3 \end{pmatrix}.$$

we can also double check $AA^{-1} = I$

$$\begin{aligned}
 \text{row 1} = (0, 1, -1) : \quad & 0 - 4 + 5 = 1, & 0 - 1 + 1 = 0, & 0 + 3 - 3 = 0 \\
 \text{row 2} = (3, 1, 1) : \quad & 9 - 4 - 5 = 0, & 3 - 1 - 1 = 1, & -6 + 3 + 3 = 0 \\
 \text{row 3} = (1, 2, -1) : \quad & 3 - 8 + 5 = 0, & 1 - 2 + 1 = 0, & -2 + 6 - 3 = 1
 \end{aligned}$$

so

$$AA^{-1} = \begin{pmatrix} 0 & 1 & -1 \\ 3 & 1 & 1 \\ 1 & 2 & -1 \end{pmatrix} \begin{pmatrix} 3 & 1 & -2 \\ -4 & -1 & 3 \\ -5 & -1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I. \quad \checkmark$$

Section 1.5 #17

solve by inverting the coefficient matrix:

$$\begin{cases} x + 4y = 2 \\ 3x + 2y = -3 \end{cases} \iff \underbrace{\begin{pmatrix} 1 & 4 \\ 3 & 2 \end{pmatrix}}_A \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{\vec{x}} = \underbrace{\begin{pmatrix} 2 \\ -3 \end{pmatrix}}_{\vec{b}}$$

find A^{-1} .

$$\left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{array} \right] \xrightarrow{R_2 - 3R_1} \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & -10 & -3 & 1 \end{array} \right] \xrightarrow{-\frac{1}{10}R_2} \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & 1 & \frac{3}{10} & -\frac{1}{10} \end{array} \right]$$

$$\xrightarrow{R_1 - 4R_2} \left[\begin{array}{cc|cc} 1 & 0 & -\frac{1}{5} & \frac{2}{5} \\ 0 & 1 & \frac{3}{10} & -\frac{1}{10} \end{array} \right] \implies A^{-1} = \begin{pmatrix} -\frac{1}{5} & \frac{2}{5} \\ \frac{3}{10} & -\frac{1}{10} \end{pmatrix}.$$

then $\vec{x} = A^{-1}\vec{b}$:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -\frac{1}{5} & \frac{2}{5} \\ \frac{3}{10} & -\frac{1}{10} \end{pmatrix} \begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} -\frac{2}{5} - \frac{6}{5} \\ \frac{6}{10} + \frac{3}{10} \end{pmatrix} = \begin{pmatrix} -\frac{8}{5} \\ \frac{9}{10} \end{pmatrix}$$

so $x = -\frac{8}{5}$ and $y = \frac{9}{10}$.

and we can also check the original system,

$$x + 4y = -\frac{16}{10} + \frac{36}{10} = 2, \quad 3x + 2y = -\frac{48}{10} + \frac{18}{10} = -3. \quad \checkmark$$

Section 1.5 #29

Suppose A is an $n \times n$ matrix and $\vec{u}, \vec{v} \in \mathbb{R}^n$. Show that if $A\vec{u} = A\vec{v}$ and $\vec{u} \neq \vec{v}$, then A is not invertible.

lets say that A actually *is* invertible, so A^{-1} exists. multiplying both sides of $A\vec{u} = A\vec{v}$ on the left by A^{-1} ,

$$\begin{aligned} A^{-1}(A\vec{u}) &= A^{-1}(A\vec{v}) \\ (A^{-1}A)\vec{u} &= (A^{-1}A)\vec{v} \\ I\vec{u} &= I\vec{v} \\ \vec{u} &= \vec{v}. \end{aligned}$$

but $\vec{u} \neq \vec{v}$ was given, so this is a contradiction. so this assumption that A^{-1} exists is false and A is not invertible.